SPOT-DETR: Tiny object detection in aerial images using high-resolution feature refinement and off-grid candidate memory

Ki-won Eom[1†], Cheoneum Park[1†], Eunkyung Kim[2*], Haneol Jang[1*]

[1]Department of Computer Engineering, Hanbat National University, 125 Dongseo-daero, Yuseong-gu, Daejeon, 34158, Republic of Korea.
[2]Department of Artificial Intelligence Software, Hanbat National University, 109 Jiphyeonbuk-ro, Sejong, 30139, Republic of Korea.

*Corresponding author(s). E-mail(s): ekim@hanbat.ac.kr; hejang@hanbat.ac.kr;
†These authors contributed equally to this work.

Abstract

Tiny object detection in remote sensing images remains difficult for two reasons: repeated downsampling in the backbone erases the fine details of tiny objects, and initial candidates fixed at cell centers are easily misaligned with object centers, so an error of a few pixels sharply lowers the intersection over union (IoU) of small boxes. We propose the Spatially-gated Path with Off-grid Tokens (SPOT), which performs feature refinement and off-grid query initialization on a single high-resolution feature path, and build the end-to-end detector SPOT-DETR with it. SPOT adds the stride-4 backbone feature to the feature pyramid and refines it with the Laplacian-based spatial gate of the Spectral-Guided Selector (SGS). The Adaptive Tiny-Object Off-Grid Memory (ATOM) then generates, in each 2×2 block of the fused feature, atoms with an occupancy logit, continuous center coordinates and a box size; these off-grid candidates compete with grid candidates to serve as initial decoder queries and complement the feature through local rasterization. Under a common training protocol, SPOT-DETR achieved an average precision (AP) of 37.6% on AI-TOD and 33.3% on AI-TOD-v2, the best among the compared detectors, with the largest gain over the SMWG-DETR baseline on AI-TOD-v2 (5.1%) for very tiny objects.

Keywords: object detection, computer vision, tiny object detection, remote sensing, aerial images, detection transformer, high-resolution features, query initialization

1 Introduction

Tiny object detection is an important task in remote sensing image interpretation, with applications in fields such as traffic management, maritime surveillance, disaster response and urban planning [1, 2]. Because tiny objects are represented by very few pixels, they carry little shape and texture information and have low contrast against the background. Combined with the wide coverage, complex backgrounds and uneven object density characteristic of remote sensing images, these properties cause general-purpose detectors to perform markedly worse on tiny objects [1, 2].

By architecture, object detectors can be divided into convolutional neural network (CNN)-based and Transformer-based detectors. CNN-based detectors have steadily advanced tiny object detection in remote sensing images through feature pyramids, selective processing of regions of interest and cluster-based re-detection [3–6]. The Detection Transformer (DETR), in turn, formulated object detection as a set prediction problem and enabled end-to-end detection without anchors or non-maximum suppression (NMS) [7]. As subsequent methods such as Deformable DETR, DAB-DETR, DN-DETR and DINO improved convergence speed and query initialization, DETR-based detectors came to surpass CNN-based detectors in generic object detection [8–11]. More recently, DETR-based detectors specialized for tiny objects, such as DQ-DETR [12], SMWG-DETR [13] and Dome-DETR [14], have been proposed. However, these methods still have two limitations, one in feature representation and one in query initialization.

First, tiny objects depend on the fine spatial information of low-level features, but cues on boundaries, local contrast and center locations fade quickly during repeated downsampling in the backbone. Deformable DETR [8], DINO [11] and RT-DETR [15] use only features with a stride of 8 or more in their default settings, and a recent DETR-based detector for tiny objects that strengthens feature interaction also takes only these features as input [13]. With a stride of 8, a very tiny object whose side is 8 pixels or shorter is represented by only one or two cells even in the finest feature map and is easily mixed with the responses of adjacent objects or the background (Fig. 1). In our experiments, the top 900 initial candidates of such a detector covered only 38.5% of very tiny objects with an IoU of 0.5 or higher, far below the 84.5% for tiny objects (Section 4.3.2). Simply adding high-resolution features, however, also brings in background responses from roads, buildings and ground textures along with the fine details [3, 4]. A high-resolution feature path is therefore needed that preserves fine details and can modulate them position by position.

Second, recent DETR-based detectors, including DINO, generate candidates at each grid cell of the encoder features and use the top candidates as the initial queries and reference boxes of the decoder [11]. Because each candidate starts from an anchor fixed at a cell center, the offset between the cell center and the actual object center must be corrected by regression alone. This localization error is particularly critical for tiny objects. For example, if the predicted center of an 8×8-pixel object is off by only 2 pixels both horizontally and vertically, the IoU drops to about 0.39, below the IoU threshold of 0.5. Initial candidates that start from the cell centers of a stride-8 grid are misaligned with the ground-truth centers (Fig. 1), and the median center error

remained 1.49 pixels even after regression (Section 4.3.2). DQ-DETR [12] and Dome-DETR [14] adjust the number and spatial distribution of queries according to the number and density of objects, but the initial position of each query is still determined at the cell level. Moreover, because candidate quality depends on the features from which the candidates are generated, changing only the query generation scheme is unlikely to yield reliable initial candidates when those features do not retain enough object cues; the two limitations therefore need to be addressed together.

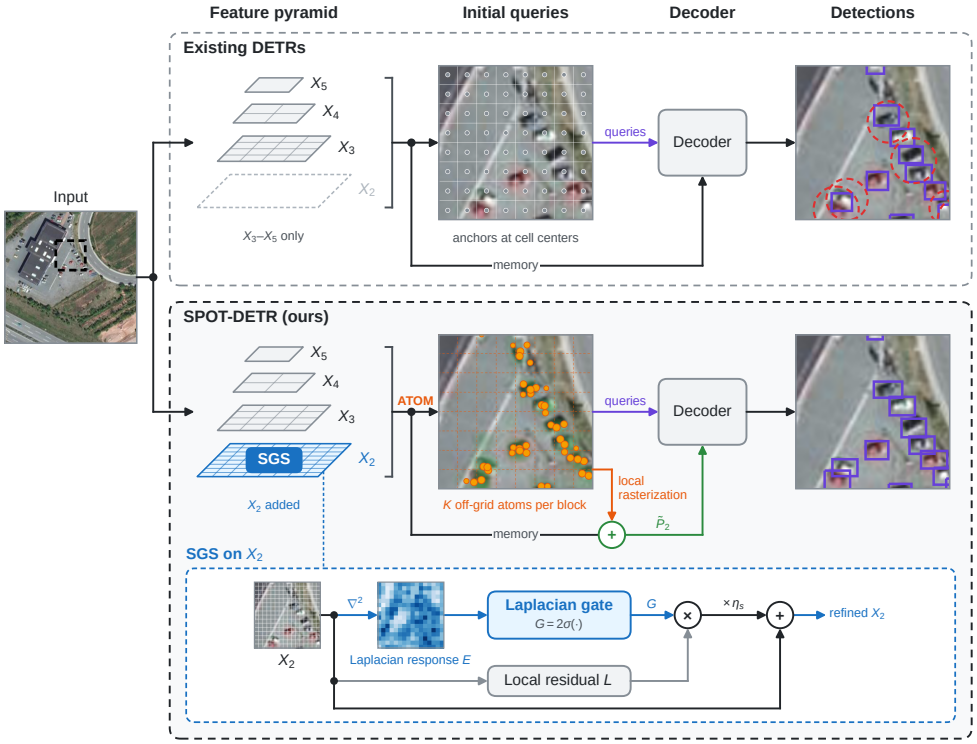


Fig. 1 Pipeline comparison between existing DETR-based detectors and SPOT-DETR. Both pipelines take the same AI-TOD-v2 test image as input, and the initial-query and detection panels magnify the dashed square region (64×64 pixels) of the input. White circles are anchors at the cell centers of a stride-8 grid, orange dots are atoms, orange dashed lines are the 2×2 block boundaries of P_2, and green indicates local rasterization. Purple boxes are predictions, and red dashed circles mark objects missed by the existing detector. The inset at the bottom shows how SGS operates on X_2. In the inset, the X_2 panel overlays the X_2 grid on the image of the same region, and E is the response computed by the trained SPOT-DETR.

Based on this analysis, we propose SPOT, which performs feature refinement and off-grid query initialization on a single high-resolution feature path, and build SPOT-DETR, an end-to-end tiny object detector that applies SPOT to SMWG-DETR. SPOT consists of two stages, each addressing one of the two limitations above. The first stage includes X_2, the highest-resolution backbone feature, in the feature pyramid

and refines it with SGS (Fig. 1). SGS generates a spatial gate from local structural cues extracted with a Laplacian filter and multiplies the gate with a local residual obtained by a lightweight local transformation. This process is designed to adjust, position by position, how much of the local residual is applied, without changing the feature resolution. The refined X_2 is fused with the semantic features of higher stages into P_2, which contains both the fine details of tiny objects and multi-scale context.

In the second stage, ATOM is applied to the fused P_2. ATOM generates atoms in each 2×2 block of P_2, and each atom consists of an occupancy logit, continuous center coordinates, a box size, a raster support and an atom feature. As off-grid candidates that are not restricted to cell centers, the atoms form the initial queries and reference boxes of the decoder together with the grid candidates of higher stages. They are also written back to P_2 through local rasterization to complement the decoder memory (Fig. 1).

In experiments under a common training protocol (Section 4.1.4) on the AI-TOD and AI-TOD-v2 datasets, SPOT-DETR achieved an AP of 37.6% and 33.3%, respectively, reaching state-of-the-art (SOTA) performance. On AI-TOD-v2, its AP and its AP for very tiny objects (APvt) were 3.5% and 5.1% higher, respectively, than those of the SMWG-DETR baseline. In the analysis of initial candidates, the candidate recall for very tiny objects rose from 38.5% to 62.8% when X_2 was included and to 67.1% in SPOT-DETR, which uses atom candidates, and the median center error of the atoms was 1.00 pixel, smaller than that of the grid candidates of the baseline (1.49 pixels). The main contributions of this study are as follows.

1. We quantitatively analyze two factors that limit the tiny object detection performance of DETR-based detectors: the loss of fine details caused by downsampling and the localization error of initial candidates fixed at cell centers. We show that, in a DETR-based detector that uses only features with a stride of 8 or more, the initial candidates miss most very tiny objects and remain offset from object centers even after regression.
2. We propose SPOT, which addresses both factors on a single high-resolution feature path. SPOT consists of SGS, which refines X_2 with a Laplacian-based spatial gate, and ATOM, which generates atoms that are not restricted to cell centers and uses them for both off-grid query initialization and feature complementation.
3. We present SPOT-DETR, a detector equipped with SPOT, compare it with existing detectors on the AI-TOD and AI-TOD-v2 datasets and analyze its components on AI-TOD-v2. SPOT-DETR achieved a higher AP than all detectors compared under the same protocol.

2 Related work

2.1 Tiny object detection in remote sensing images

Tiny object detection is the problem of finding the locations and categories of objects that are represented by very few pixels in an image; in remote sensing images, objects whose bounding-box area is smaller than 16×16 pixels are generally regarded as tiny objects [1, 2]. Because a remote sensing image covers a wide area in a single scene,

many of the objects of practical interest, such as vehicles, ships, people and storage tanks, appear at this size. In applications where the number and locations of individual objects are themselves the result of the analysis, such as traffic flow analysis and ship monitoring, missing tiny objects distorts the subsequent counting, tracking and situational assessment. The performance of tiny object detection is therefore directly tied to the reliability of wide-area remote sensing image interpretation.

However, most generic object detectors are designed for images in which objects appear relatively large, and they struggle to achieve satisfactory performance when applied directly to tiny objects [1, 2]. The causes can be summarized in three points. First, after downsampling in the backbone, the boundary and texture information of tiny objects shrinks to one or two cells or disappears in deep feature maps. Second, IoU-based label assignment struggles to secure enough positive samples reliably for tiny objects, whose IoU changes sharply with a positional difference of a few pixels [2, 16, 17]. Third, object-like responses arising from the wide background easily overwhelm weak object responses. In response, research on tiny object detection has developed in three directions: region selection, the redesign of label assignment and location similarity measures, and the enhancement of feature representations.

Region selection methods concentrate computation on regions of a wide-area image that are likely to contain objects. ClusDet predicted clustered regions and the object scale within them and then detected objects in the selected regions after normalizing them [5]. DMNet cropped regions crowded with objects using a density map and detected objects in them again [6]. These methods can distribute computation efficiently over wide-area images, but they may miss sparse objects when the preceding cluster or density estimation is inaccurate, and they must merge the results of multiple regions.

Studies on label assignment and location similarity measures focus on alleviating the positional sensitivity of IoU. Generalized IoU (GIoU) used the smallest region enclosing both boxes so that the positional difference was reflected even when the two boxes did not overlap [18]. Dot distance replaced IoU with a measure that normalizes the distance between two box centers by the average object size of the dataset [16], and the normalized Wasserstein distance (NWD) modeled bounding boxes as two-dimensional Gaussian distributions and measured their similarity with the Wasserstein distance [2]. Gaussian receptive field based label assignment (RFLA) modeled the receptive field of features as a Gaussian and assigned positive samples hierarchically according to their distance from the ground truth [17]. However, these measures were designed mainly for label assignment in CNN-based detectors and depend on specific bounding-box representations or on reference values that must be tuned for each dataset. Instead of designing a new measure, we improve the query initialization stage, where anchors fixed at cell centers introduce an initial offset from object centers.

2.2 CNN-based tiny object detection

CNN-based detectors have mainly developed toward reinforcing the weak feature representations of tiny objects. The Feature Pyramid Network (FPN) combined the location information of high-resolution features with the semantic information of deep features through a top-down pathway and lateral connections, laying the foundation for

multi-scale detection [3]. QueryDet predicted the coarse locations of small objects on low-resolution features and then sparsely processed only the high-resolution features at those locations, improving the balance between fine details and computational cost [4]. CFINet produced high-quality proposals for small objects through coarse-to-fine proposal generation consisting of dynamic anchor selection and cascaded regression, and reinforced their representations through feature imitation learning [19].

Many CNN-based detectors have also been proposed for tiny objects in remote sensing images. DNTR combined a denoising FPN with a self-attention-based Trans R-CNN [20]. ORFENet reinforced the features of tiny objects through object reconstruction and adaptive feature enhancement with multiple receptive fields [21], and LTDNet improved detection efficiency by exploring lightweight structures [22]. More recently, CMA-Net combined channel and foreground attention with a weighted bidirectional pyramid [23], a dynamic element-activated method used local sparse attention and cross-level channel interaction together with diffusion wavelet convolution [24], and Aero-LiteNet combined multi-resolution fusion and cross-dimensional attention with a localization loss that accounts for the neighborhood relations of dense objects [25].

These methods showed that high-resolution features and multi-scale fusion are important for tiny object detection, but most of them rely on dense prediction and hand-crafted post-processing such as NMS. In addition, adding high-resolution stages or repeating feature fusion can bring in background responses along with the fine details of objects and can increase computational cost [3, 4]. SPOT also exploits high-resolution features, but it differs in that it includes the X_2 feature in the feature pyramid and decoder memory of an end-to-end DETR, modulates its local residual with SGS and carries this feature through to query initialization.

2.3 DETR-based tiny object detection

DETR formulated object detection as the direct prediction of a fixed-size set and removed the dependence on anchors and NMS by assigning each ground truth one-to-one to a prediction through Hungarian matching [7]. Subsequent studies have focused on convergence speed and query design. Deformable DETR introduced multi-scale deformable attention, which samples only a few locations around each reference point [8]. DAB-DETR used bounding-box coordinates as the positional component of queries [9], and DN-DETR reduced the instability of matching with an auxiliary task that reconstructs noised ground-truth boxes [10]. DINO introduced contrastive denoising training together with mixed query selection, which initializes the anchor boxes of the decoder with the positions of the top-ranked features selected from the encoder [11]. RT-DETR designed a hybrid encoder that processes the features of the last three backbone stages and selected initial queries from the encoder features, achieving real-time end-to-end detection [15]. In DINO and RT-DETR, the positions of the initial queries are thus determined by the cells of the encoder feature grid.

DETR research for tiny objects has expanded toward reflecting the number, density and features of objects in the construction of queries. DQ-DETR adjusted the number of queries and their positional information for each image using category-wise counting and a density map, and used the same signals to enhance the features of tiny

objects [12]. Dome-DETR concentrated attention computation on foreground regions using a density map estimated from shallow features, and progressively adjusted the number and placement of queries according to the density [14]. Built on RT-DETR, SMWG-DETR designed an adaptive spectral modulation module and a wavelet-guided fusion module to improve single-scale feature interaction and multi-scale fusion with frequency information, and refined the supervision of contrastive denoising training with dynamic denoising query selection [13].

Although these studies improved the number and distribution of queries, feature representations and training supervision, the initial position of each query is still determined at the cell level. In contrast, ATOM creates off-grid candidates by predicting continuous center coordinates and occupancy logits within each block under a fixed number of queries, and uses the same atoms to complement the features.

3 Methods

3.1 Overall architecture

Figure 2 shows the overall architecture of SPOT-DETR. Features with strides of 4, 8, 16 and 32 are extracted from a ResNet-50 [26] backbone and converted into the four-level 256-channel features X_2–X_5. X_2, the feature with the highest resolution, is refined by SGS, and X_3–X_5 undergo single-scale feature interaction through spectral modulation [13]. Without a separate Transformer encoder, the four-level features pass through wavelet-guided top-down fusion and bottom-up path aggregation to become the multi-scale features P_2–P_5 [13]. ATOM then generates atoms with continuous center coordinates from the fused P_2 and outputs atom candidates together with $\widetilde{P}_2$, which is complemented by local rasterization. The top 900 are selected from the union of the atom candidates and the grid candidates of P_3–P_5 to form the initial queries and reference boxes of the decoder, and $\widetilde{P}_2$ and P_3–P_5 serve as the decoder memory. A six-layer deformable Transformer decoder [8] iteratively updates the initial queries and predicts a category score and a bounding box for each query. During training, predictions are matched to ground truths by Hungarian matching, and contrastive denoising training with denoising query selection is applied in parallel [7, 11, 13]. At inference, the output of the last decoder layer is used directly without NMS [7, 11].

Within this architecture, SPOT is the high-resolution feature path from X_2 through SGS and multi-scale fusion to ATOM; SGS and ATOM are the parts newly designed in this study. Spectral modulation, wavelet-guided fusion and denoising training follow the implementation of [13], with the fusion extended only to include X_2, so that the effect of SPOT can be evaluated in isolation. The overall training objective is summarized at the end of Section 3.3.

3.2 SGS

Cues on the boundaries and center locations of tiny objects remain mainly in the shallow stages of the backbone. The first stage of SPOT includes X_2, which carries these cues, in the feature pyramid and, before fusion, refines it with SGS, which is

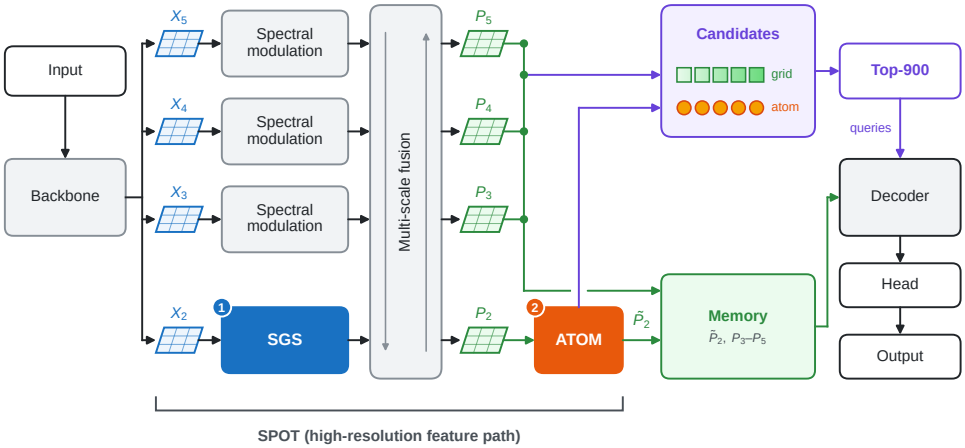


Fig. 2 Overall architecture of SPOT-DETR. Of the ResNet-50 features X_2–X_5, X_2 is refined by SGS and X_3–X_5 are processed by spectral modulation, after which multi-scale fusion produces P_2–P_5. ATOM generates atoms from P_2 and outputs a raster residual and atom candidates. The entire high-resolution feature path from X_2 through SGS, fusion and ATOM constitutes SPOT. The top 900 of the atom candidates and the P_3–P_5 grid candidates initialize the decoder, and $\widetilde{P}_2, P_3, P_4, P_5$ are used as the decoder memory.

designed to modulate a local residual position by position. After multi-scale fusion, the refined X_2 becomes the P_2 used by ATOM (Section 3.3).

High-resolution feature path. Let the outputs of the four backbone stages be C_l ($l \in \{2,3,4,5\}$), so that the stride of each feature is $s_l = 2^l$. Each C_l is projected to 256 channels by a 1×1 convolution and batch normalization (BN):

$$X_l = \mathrm{BN}(\mathrm{Conv}_{1\times1}(C_l)), \qquad X_l \in \mathbb{R}^{256\times H/s_l\times W/s_l} \tag{1}$$

For the 800×800 input used in the experiments, the spatial sizes of X_2–X_5 are 200×200, 100×100, 50×50 and 25×25, respectively. Adding X_2 to X_3–X_5 thus retains in the feature pyramid location and boundary cues finer than those available at the higher stages.

Instead of being fused immediately, X_2 is first refined by two SGS blocks, while two spectral modulation blocks are applied to each of X_3–X_5 [13]. The four-level features then undergo three wavelet-guided top-down fusions. Each fusion uses the Haar wavelet components of the finer level to upsample the higher-level feature in a content-adaptive manner and to complement detailed responses [13]. Information is then re-aggregated through three stride-2 downsamplings and CSPRep fusion [15], and a final 1×1 projection outputs the 256-channel P_2, P_3, P_4, P_5.

SGS block. SGS is a token mixer that generates a spatial gate from high-frequency local structural cues extracted with a Laplacian filter and multiplies the gate with a local residual (Fig. 3). Because the Laplacian is a high-pass filter, the spatial gate is derived from the high-frequency components of the feature, which correspond to boundaries and local contrast. The SGS block follows the pre-normalization residual

structure of MetaFormer [27, 28]; for an input feature X,

$$Z = \mathrm{LN}(X), \qquad U = X + \mathcal{S}(Z), \qquad Y = U + \mathrm{MLP}(\mathrm{LN}(U)) \tag{2}$$

where $\mathcal{S}$ is the token-mixing operation of SGS, LN denotes layer normalization and MLP denotes a multilayer perceptron. The local residual $L(Z)$ is obtained by reducing the number of channels of Z from 256 to $d_s = 64$, applying a 3×3 depthwise convolution–BN–sigmoid linear unit (SiLU) and a 1×1 pointwise convolution–BN–SiLU, and projecting the result back to 256 channels. The channel MLP in Eq. (2) expands the 256 channels to 1,024, applies StarReLU [28] and restores the original dimension.

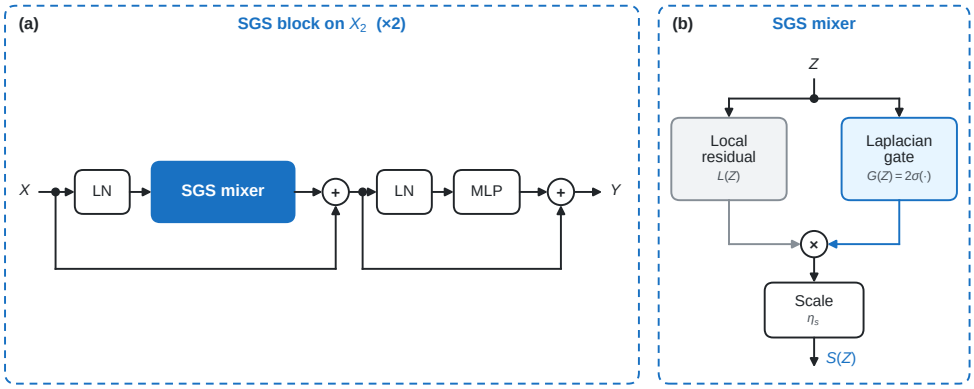


Fig. 3 Structure of SGS. (a) SGS block, applied twice in succession to X_2 before fusion. (b) Token-mixing operation $\mathcal{S}$ of SGS.

Laplacian-based spatial gate. The spatial gate is generated from the Laplacian response of the normalized feature Z. A fixed kernel K_{lap} is applied to each channel with zero padding of 1 to obtain the channel-averaged Laplacian response map E:

$$K_{\mathrm{lap}} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}, \qquad E_{hw} = \frac{1}{256} \sum_{c=1}^{256} \left| (K_{\mathrm{lap}} * Z_c)_{hw} \right| \tag{3}$$

E passes through a 16-channel 3×3 convolution, SiLU and a 1×1 convolution in sequence, and the spatial gate G and the token-mixing output of SGS are computed as

$$G = 2\,\sigma\big(\mathrm{Conv}_{1\times1}\big(\mathrm{SiLU}(\mathrm{Conv}_{3\times3}(E))\big)\big), \qquad \mathcal{S}(Z) = \eta_s\, G \odot L(Z) \tag{4}$$

The single-channel map G is multiplied identically with all 256 feature channels. The weights and bias of the last gate convolution are initialized to zero so that $G = 1$ at the beginning of training, and the learnable per-block residual coefficient η_s is initialized to 0.1. Through Eq. (4), SGS can therefore adjust, position by position, how much of

the local residual is applied while keeping the spatial resolution and the number of tokens of X_2.

3.3 ATOM

As analyzed in Section 1, initial candidates that start from cell centers begin regression misaligned with the centers of tiny objects, and for objects a few pixels in size this error directly lowers the IoU. ATOM, the second stage of SPOT, generates atoms, which are block-level object representations, from the fused P_2 and uses them for both feature complementation and query initialization (Fig. 4). Here, off-grid means that the continuous center coordinates of an atom are not fixed at a cell center but can take any position inside the block to which the atom belongs. Atom candidates compete with the grid candidates of higher stages for selection as initial queries and reference boxes, while $\widetilde{P}_2$, which is complemented by the raster residual, is used together with the original P_3–P_5 as the decoder memory.

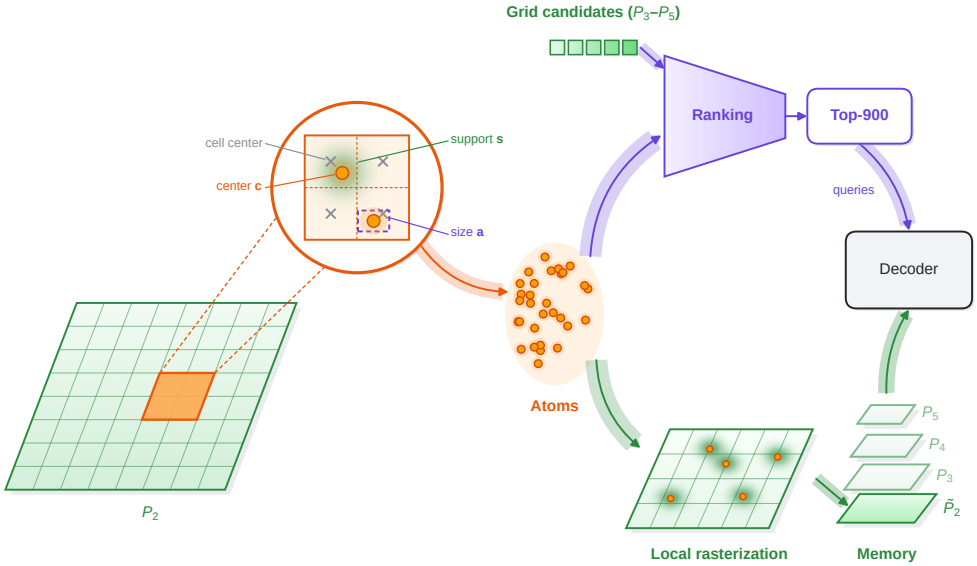


Fig. 4 Structure of ATOM. Atoms generated in each 2×2 block of the fused P_2 produce $\widetilde{P}_2$ through local rasterization and are ranked together with the grid candidates of P_3–P_5 to form the top 900 initial queries.

Atom generation. ATOM reduces $P_2 \in \mathbb{R}^{256\times H_2\times W_2}$ to a $d_a = 64$-channel feature Z_a with a 1×1 convolution, group normalization and SiLU. Z_a is divided into non-overlapping $B \times B$ blocks, and we use $B = 2$ and $K = 2$ atoms per block. Let $\mathbf{z}_{rj} \in \mathbb{R}^{64}$ $(j = 1, \ldots, B^2)$ be the cell features of block r, and let $\mathbf{h}_r \in \mathbb{R}^{128}$ be the representation obtained by concatenating the four cell features and transforming them with an MLP. With the affine prediction layers $\mathcal{H}_o, \mathcal{H}_c, \mathcal{H}_s, \mathcal{H}_a, \mathcal{H}_\beta, \mathcal{H}_v$, which include

biases, the atom parameters of a block are computed as

$$
\begin{aligned}
\mathbf{o}_r &= \mathcal{H}_o(\mathbf{h}_r) \in \mathbb{R}^K, \\
\mathbf{C}_r &= \mathbf{1}_K \boldsymbol{\pi}_r^{\mathsf{T}} + B\,\sigma\big(\mathrm{reshape}_{K\times 2}\,\mathcal{H}_c(\mathbf{h}_r)\big), \\
\mathbf{S}_r &= 0.35 + 1.15\,\sigma\big(\mathrm{reshape}_{K\times 2}\,\mathcal{H}_s(\mathbf{h}_r)\big), \\
\mathbf{A}_r &= 0.50 + 3.50\,\sigma\big(\mathrm{reshape}_{K\times 2}\,\mathcal{H}_a(\mathbf{h}_r)\big), \\
\boldsymbol{\beta}_r &= \mathrm{softmax}_j\big(\mathrm{reshape}_{K\times B^2}\,\mathcal{H}_\beta(\mathbf{h}_r)\big), \qquad \mathbf{v}_{rk} = \mathcal{H}_v\left(\sum_{j=1}^{B^2} \beta_{rkj}\mathbf{z}_{rj}\right)
\end{aligned}
\tag{5}
$$

where $\boldsymbol{\pi}_r$ is the top-left coordinate of the block and $\mathbf{v}_{rk} \in \mathbb{R}^{64}$ is the atom feature. We denote the kth element of $\mathbf{o}_r$ by o_{rk} and the kth rows of $\mathbf{C}_r, \mathbf{S}_r, \mathbf{A}_r$ by $\mathbf{c}_{rk}, \mathbf{s}_{rk}, \mathbf{a}_{rk}$, respectively. The continuous center coordinates are restricted to the inside of the corresponding 2×2 block, and the raster support and box size range over $[0.35, 1.50]$ and $[0.50, 4.00]$, respectively, in units of the P_2 grid. To reflect the rarity of objects, the bias of the occupancy prediction layer is initialized to -2, and the biases of the center prediction layer are set differently for the two atoms so that their initial centers do not overlap. Although the number of atoms per block is fixed, the continuous center coordinates, box size, raster support, occupancy logit and atom feature of each atom vary with the input.

The atom feature is converted into a 256-dimensional query feature through a skip projection and a residual MLP. The continuous center coordinates and box size are normalized by the width and height of P_2 and clipped to $[10^{-5}, 1-10^{-5}]$ to form the base box $\mathbf{b}_i^0$. The projection of the query feature and the classification and regression branches are computed as

$$
\begin{aligned}
\mathbf{q}_i &= W_{\mathrm{skip}}\mathbf{v}_i + \tanh(g_q)\,\mathrm{MLP}(\mathbf{v}_i), \qquad \mathbf{m}_i = \mathrm{LN}(\mathcal{H}_m(\mathbf{q}_i)), \\
\mathbf{b}_i^0 &= \mathrm{clip}\left[\left(\frac{c_{ix}}{W_2}, \frac{c_{iy}}{H_2}, \frac{a_{iw}}{W_2}, \frac{a_{ih}}{H_2}\right), 10^{-5}, 1-10^{-5}\right], \\
\boldsymbol{\ell}_i^{\mathrm{atom}} &= f_{\mathrm{cls}}(\mathbf{m}_i), \qquad \overline{\mathbf{b}}_i = \sigma\big(f_{\mathrm{reg}}(\mathbf{m}_i) + \mathrm{logit}(\mathbf{b}_i^0)\big)
\end{aligned}
\tag{6}
$$

where $\mathcal{H}_m$ is an affine projection that maps the query feature into the candidate feature space of the decoder. For selected atom candidates, $\mathbf{m}_i$ and $\overline{\mathbf{b}}_i$ become the initial query and the reference box of the decoder, respectively. During training, $\mathbf{m}_i$ and $\overline{\mathbf{b}}_i$ are passed to the decoder with their gradients stopped, and the initial-candidate loss is applied directly to $(\boldsymbol{\ell}_i^{\mathrm{atom}}, \overline{\mathbf{b}}_i)$. The residual gate g_q is initialized to zero.

Feature complementation through local rasterization. All atoms before the top-900 selection take part in local rasterization. Each atom feature is accumulated with weights only into the neighborhood of at most 3×3 cells around its center cell. With Ω denoting the set of valid P_2 cell centers, the neighbor set $\mathcal{N}_i$ of atom i, the

raster weight κ_i, the raster feature R and the complemented feature $\widetilde{P}_2$ are defined as

$$\begin{aligned}
\mathcal{N}_i &= \left\{ \lfloor \mathbf{c}_i \rfloor + \boldsymbol{\delta} + (1/2, 1/2) \mid \boldsymbol{\delta} \in \{-1, 0, 1\}^2 \right\} \cap \Omega, \\
\kappa_i(\mathbf{u}) &= \mathbf{1}[\mathbf{u} \in \mathcal{N}_i] \exp\left[-\frac{1}{2} \sum_{d \in \{x,y\}} \left(\frac{u_d - c_{id}}{s_{id}} \right)^2 \right] \sigma(o_i), \\
R(\mathbf{u}) &= \frac{\sum_i \kappa_i(\mathbf{u}) W_r \mathbf{v}_i}{\max(\sum_i \kappa_i(\mathbf{u}), 10^{-6})}, \qquad \widetilde{P}_2 = P_2 + \rho_e \tanh(g_r) W_{\text{out}} R
\end{aligned} \tag{7}$$

The intersection with Ω excludes cells outside the image, and the indicator function $\mathbf{1}[\cdot]$ limits the contribution of each atom to its 3×3 neighborhood. g_r is a learnable gate initialized to zero, and ρ_e is a coefficient that sets how much of the raster residual is applied at each stage of training. The normalization by the denominator prevents the feature magnitude from growing cumulatively when several atoms contribute to one cell.

Mixed candidate selection. With ℓ_{ic}^{atom} and ℓ_{ic}^{grid} denoting the classification logits of atom and grid candidates, respectively, the candidates are ranked by

$$r_i^{\text{atom}} = \max_c \ell_{ic}^{\text{atom}} + \lambda_{\text{rank}} \operatorname{stopgrad}(o_i), \qquad \lambda_{\text{rank}} = 0.25, \qquad r_i^{\text{grid}} = \max_c \ell_{ic}^{\text{grid}} \tag{8}$$

The occupancy logit is added only to the ranking score, and no gradient flows through this path. Hence, λ_{rank} affects only the candidate ranking and is not part of the loss function. SPOT-DETR replaces the grid candidates of P_2 with atom candidates and keeps the grid candidates of P_3–P_5. Because the top 900 are selected from the union of the two candidate sets, the proportion of selected atom candidates varies from image to image. For an 800×800 input, 20,000 atom candidates are generated from 10,000 blocks and compete with 13,125 grid candidates of P_3–P_5. The decoder memory, in contrast, retains all 53,125 tokens of $\widetilde{P}_2$ and P_3–P_5.

Auxiliary assignment and supervision. The auxiliary location and size supervision of ATOM is applied only to objects whose bounding box has a longer side of 16 pixels or less. Ground-truth centers are converted into P_2 grid coordinates and grouped by the 2×2 block that contains them. In each block, bipartite matching is performed so that the sum of the L_1 distances between the gradient-stopped predicted centers and the ground-truth centers is minimized. The size target is defined as $\text{clamp}((w, h)/4, 0.5, 4.0)$. If a block contains more than K target objects, only the first K in annotation order are used for auxiliary supervision. The main detection loss is applied to all ground truths regardless of object size.

The auxiliary objective of ATOM is defined as

$$\mathcal{L}_{\text{ATOM}} = \mathcal{L}_{\text{occ}} + 2\mathcal{L}_{\text{ctr}} + 0.5\mathcal{L}_{\text{size}} + 0.25\mathcal{L}_{\text{cycle}} + 0.1\mathcal{L}_{\text{count}} \tag{9}$$

$\mathcal{L}_{\text{occ}}$ is the sum of the mean binary cross-entropy over positive atoms and over valid negative atoms, with the negative term multiplied by 0.25. Smooth-L_1 is applied to the centers of matched atoms and Smooth-L_1 in log space to their sizes. With M_r denoting the number of target objects in block r, $\mathcal{L}_{\text{count}}$ is the Smooth-L_1 loss between

the sum of the occupancy probabilities in the block and $\min(M_r, K)$. $\mathcal{L}_{\text{cycle}}$ encourages the raster feature to reconstruct the gradient-stopped 64-channel reduced P_2 feature in target object regions and imposes a suppression term with a weight of 0.1 on the mean absolute value of the raster feature in background regions. Blocks that contain only the centers of objects exceeding the 16-pixel criterion are excluded from the computation of $\mathcal{L}_{\text{occ}}$ and $\mathcal{L}_{\text{count}}$.

Staged training. To prevent atom candidates, which are still unstable early in training, from disturbing the training of the decoder, the same staged schedule is applied to the atom candidate allocation ratio q_e and the raster residual coefficient ρ_e. With the epoch index e starting from 1, the schedule is

$$q_e = \rho_e = \begin{cases} 0, & e \le 5, \\ 0.25, & 6 \le e \le 10, \\ 0.25 + 0.75(e-10)/11, & 11 \le e \le 20, \\ 1, & e \ge 21 \end{cases} \tag{10}$$

During epochs 1–5, only the grid candidates of P_2–P_5 are used and the raster residual is not applied, but $\mathcal{L}_{\text{ATOM}}$ is included in the total objective from the first epoch. The two coefficients increase linearly during epochs 11–20 and become 1 from epoch 21. For $0 < q_e < 1$, $\text{round}(900q_e)$ slots are allocated to atom candidates, and the rest are selected from the grid candidates of P_2–P_5. Once $q_e = 1$, the top 900 are selected from the union of the atom candidates and the grid candidates of P_3–P_5. The final architecture with $q_e = \rho_e = 1$ is used for validation and inference.

Overall training objective. SPOT-DETR jointly learns classification, bounding-box regression and the auxiliary supervision of ATOM end to end. For bounding-box regression and matching, we do not introduce a location similarity measure for tiny objects such as NWD [2] but keep the L_1–GIoU configuration of [13], so the only newly added loss is the auxiliary loss of ATOM. The Hungarian matching cost consists of a focal classification cost [29], an L_1 box cost and a GIoU cost [18], and the cost between prediction i and ground truth j is defined as

$$\mathcal{C}_{ij} = 2\mathcal{C}_{\text{focal}}(i,j) + 5\left\|\widehat{\mathbf{b}}_i - \mathbf{b}_j\right\|_1 - 2\,\text{GIoU}(\widehat{\mathbf{b}}_i, \mathbf{b}_j) \tag{11}$$

where $\widehat{\mathbf{b}}_i$ and $\mathbf{b}_j$ are normalized (c_x, c_y, w, h) coordinates, and $\alpha = 0.25$ and $\gamma = 2$ are used for the focal classification cost. The detection objective of each decoder layer and of the initial-candidate output is

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{VFL}} + 5\mathcal{L}_1 + 2\mathcal{L}_{\text{GIoU}} \tag{12}$$

with $\mathcal{L}_{\text{GIoU}} = 1 - \text{GIoU}$. $\mathcal{L}_{\text{VFL}}$ is applied to all queries with the category targets of unmatched queries set to zero, whereas the L_1 and GIoU losses are applied only to matched predictions. The varifocal loss uses a weight of 1, $\alpha = 0.75$ and $\gamma = 2$, and the quality target of positive samples is computed as the IoU [30].

The total objective, which sums the losses of the matched-query and denoising-query outputs of the six decoder layers, the loss of the initial-candidate output and

the auxiliary loss of ATOM, is

$$\mathcal{L}_{\text{total}} = \sum_{d=1}^{6} \left(\mathcal{L}_{\text{det},d}^{\text{match}} + \mathcal{L}_{\text{det},d}^{\text{DN}} \right) + \mathcal{L}_{\text{det}}^{\text{prop}} + \mathcal{L}_{\text{ATOM}} \tag{13}$$

Because the correspondence of denoising queries to the ground truths is determined in advance, they do not go through the Hungarian matching of Eq. (11). Contrastive denoising training and denoising query selection follow the settings of [11, 13]. Denoising query selection removes the weights of some negative classification targets that are likely to disturb training, as judged by the focal–L_1–GIoU cost, and the L_1–GIoU combination of Eq. (12) is used for all positive box regressions, including those of denoising queries and initial candidates.

4 Experiments and results

4.1 Experimental setup

4.1.1 Datasets

We used AI-TOD [1] and AI-TOD-v2 [2], benchmarks for tiny object detection in remote sensing images. AI-TOD consists of 28,036 aerial images and 700,621 objects in eight categories, with an average object size of about 12.8 pixels [1]. The categories are airplane (AI), bridge (BR), storage tank (ST), ship (SH), swimming pool (SP), vehicle (VE), person (PE) and wind mill (WM). AI-TOD-v2 re-annotates the same images to reduce the missing annotations and localization errors of AI-TOD [2]. For both datasets, models were trained on the 14,018 images of the official train and validation splits combined and evaluated on the 14,018 images of the official test split. The training and test splits of AI-TOD-v2 contain 376,625 and 376,121 annotated objects, respectively.

4.1.2 Evaluation metrics

Evaluation followed the official AI-TOD evaluation protocol [1, 2]. The main metric is the AP averaged over IoU thresholds from 0.50 to 0.95 in steps of 0.05. Performance by object size is evaluated in four ranges defined by the square root of the object area: very tiny (vt, 2–8 pixels), tiny (t, 8–16 pixels), small (s, 16–32 pixels) and medium (m, 32 pixels or more). The AP of the tiny and small ranges is denoted by APt and APs, respectively, and the comparison tables report AP, APvt, APt and APs, focusing on tiny objects. Differences in AP and recall are reported as absolute differences. In the analysis of initial candidates (Section 4.3.2), recall is defined as the fraction of ground truths that overlap at least one of the top 900 initial candidates with an IoU of 0.5 or higher. At inference, up to 900 predictions per image are output from the last decoder layer without NMS, and the maximum number of detections per image for evaluation was set to 1,500. In the qualitative figures, predictions and ground truths of the same category with an IoU of 0.5 or higher were matched in descending order of prediction score; matched predictions are marked as true positives (TP), unmatched predictions

as false positives (FP) and undetected ground truths as false negatives (FN). The score threshold of each model was set to the value that maximizes F1 over the entire AI-TOD-v2 test split.

4.1.3 Implementation details

SPOT-DETR was implemented with PyTorch [31] and MMDetection [32], and a ResNet-50 pretrained on ImageNet [33] was used as the backbone. The stem and BN parameters of the backbone were frozen during training. The model dimension is 256, and the decoder consists of six layers, eight attention heads and a 1,024-dimensional feed-forward network. The number of initial queries is 900, and the settings of contrastive denoising training and denoising query selection are the same as in [13]. The input resolution is 800×800, and no data augmentation was used. We used AdamW [34] with a learning rate and weight decay of 10^{-4} each, and the learning rate of the backbone was set to 0.1 times that value. A linear warm-up was applied for the first 2,000 iterations, and the gradient norm was clipped at 0.1. Models were trained for 40 epochs with a batch size of 4, and the exponential moving average weights were used for evaluation. All experiments were run on a single NVIDIA RTX A6000 GPU in 32-bit floating-point precision.

4.1.4 Comparison protocol

For a fair comparison, all compared models were trained for 40 epochs on the same official train and validation splits as SPOT-DETR, and their last checkpoints were evaluated on the official test split. Each model used its public implementation and default hyperparameters, and no results reported in other papers were quoted. In the comparisons with existing detectors, the performance of SPOT-DETR is reported as the mean and standard deviation over three runs with different seeds, and the ablation and analysis experiments use the model trained with seed 42. For the qualitative comparisons and feature map visualizations, SPOT-DETR used the checkpoint analyzed in Section 4.3.2, and the compared and component models used the checkpoints that produced the results in Sections 4.2 and 4.3.

4.2 Comparison with existing methods

4.2.1 Overall performance

Table 1 compares SPOT-DETR with CNN-based and DETR-based detectors on the AI-TOD-v2 test split. SPOT-DETR achieved the highest performance on all four metrics, with an AP of 33.3±0.3%, APvt of 20.8±0.3%, APt of 34.5±0.7% and APs of 36.1±0.2%. Its AP was 3.1% higher than that of DQ-DETR (30.2%), which had the second-highest AP, and its AP, APvt, APt and APs were 3.5%, 5.1%, 4.0% and 2.2% higher, respectively, than those of the SMWG-DETR baseline. The gain over the baseline was larger for smaller size ranges. Over the second-best detector in each range, the margin was 4.7% in APvt (RT-DETR, 16.1%) but only 0.4% in APs (DQ-DETR, 35.7%). The standard deviations over the three seeds (0.2–0.7%) were smaller than the gains over the baseline.

Table 1 Performance comparison on the AI-TOD-v2 test split. Values for SPOT-DETR are the mean ± standard deviation over three seeds. Red and blue indicate the best and second-best results in each column, respectively.

Method	AP	APvt	APt	APs
CNN-based				
DNTR [20]	0.215	0.089	0.216	0.256
ORFENet [21]	0.251	0.107	0.251	0.293
LTDNet [22]	0.230	0.089	0.236	0.274
DETR-based				
DINO [11]	0.279	0.150	0.282	0.327
RT-DETR [15]	0.296	0.161	0.302	0.329
RT-DETRv2 [35]	0.279	0.143	0.282	0.312
DQ-DETR [12]	0.302	0.156	0.304	0.357
Dome-DETR [14]	0.262	0.116	0.259	0.316
SMWG-DETR [13]	0.298	0.157	0.305	0.339
SPOT-DETR (ours)	0.333 ± 0.003	0.208 ± 0.003	0.345 ± 0.007	0.361 ± 0.002

The AP of SPOT-DETR was 8.2% higher than that of ORFENet (25.1%), the best CNN-based detector. Among the DETR-based detectors, the general-purpose DINO, RT-DETR and RT-DETRv2 [11, 15, 35] reached an AP of 27.9–29.6%, and DQ-DETR, Dome-DETR and SMWG-DETR [12–14], which were designed for tiny objects, reached 26.2–30.2%.

Table 2 Performance comparison on the AI-TOD test split. Values for SPOT-DETR are the mean ± standard deviation over three seeds. Red and blue indicate the best and second-best results in each column, respectively.

Method	AP	APvt	APt	APs
CNN-based				
ORFENet [21]	0.246	0.117	0.251	0.292
LTDNet [22]	0.235	0.106	0.248	0.283
DETR-based				
DQ-DETR [12]	0.310	0.171	0.315	0.368
SMWG-DETR [13]	0.321	0.176	0.332	0.372
SPOT-DETR (ours)	0.376 ± 0.001	0.233 ± 0.004	0.393 ± 0.002	0.404 ± 0.003

Table 2 shows the results on the AI-TOD test split. SPOT-DETR was the highest on all metrics, with an AP of 37.6±0.1%, APvt of 23.3±0.4%, APt of 39.3±0.2% and APs of 40.4±0.3%. Its gains over the SMWG-DETR baseline were 5.5% in AP, 5.7% in APvt, 6.1% in APt and 3.2% in APs, larger than those on AI-TOD-v2, and its AP was 6.6% higher than that of DQ-DETR. Because AI-TOD-v2 re-annotates the same

images, the consistent gains on both datasets indicate that the improvement does not depend on a particular annotation version.

4.2.2 Per-category performance

Table 3 Per-category AP comparison on the AI-TOD-v2 test split. Values for SPOT-DETR are the mean ± standard deviation over three seeds. Red and blue indicate the best and second-best results in each column, respectively.

Method	AI	BR	ST	SH	SP	VE	PE	WM
CNN-based								
DNTR [20]	0.269	0.128	0.347	0.456	0.131	0.242	0.095	0.051
ORFENet [21]	0.276	0.207	0.358	0.518	0.157	0.276	0.112	0.101
LTDNet [22]	0.234	0.200	0.344	0.450	0.175	0.273	0.096	0.072
DETR-based								
DINO [11]	0.343	0.210	0.371	0.596	0.173	0.285	0.175	0.080
RT-DETR [15]	0.375	0.224	0.358	0.628	0.198	0.316	0.179	0.087
RT-DETRv2 [35]	0.373	0.198	0.346	0.615	0.183	0.297	0.153	0.066
DQ-DETR [12]	0.404	0.233	0.425	0.461	0.230	0.327	0.184	0.152
Dome-DETR [14]	0.320	0.211	0.406	0.421	0.173	0.319	0.159	0.086
SMWG-DETR [13]	0.370	0.242	0.361	0.630	0.185	0.320	0.187	0.091
SPOT-DETR (ours)	0.389	0.263	0.406	0.665	0.210	0.372	0.240	0.122
	±0.002	±0.008	±0.001	±0.003	±0.005	±0.001	±0.003	±0.015

In the per-category results in Table 3, SPOT-DETR achieved the highest AP in four categories: BR, SH, VE and PE. In particular, its AP for VE and PE was 37.2% and 24.0%, respectively, 4.5% and 5.3% higher than that of the second-best method. DQ-DETR was the best for AI, ST, SP and WM, with differences of 1.5–3.0% from SPOT-DETR. Compared with the SMWG-DETR baseline, SPOT-DETR had a 1.9–5.3% higher AP in all eight categories, indicating that the improvement was not confined to particular categories. However, WM had a larger standard deviation across seeds (1.5%) than the other categories, so its result should be interpreted with caution.

4.2.3 Qualitative comparison

Figure 5 compares the detection results of SPOT-DETR, DQ-DETR and RT-DETR in six scenes from the AI-TOD-v2 test split. DQ-DETR has the second-highest AP after SPOT-DETR in Table 1, and RT-DETR has the second-highest APvt. In all six scenes, SPOT-DETR had the most TPs and the fewest FNs of the three models, and the differences appeared mainly where small vehicles were densely packed or lined up, as in the yellow dashed regions. For example, in a crowded parking lot scene (first row of the right group), SPOT-DETR detected 349 of the 426 ground truths, whereas DQ-DETR and RT-DETR detected 322 and 219, respectively; in a dark road scene (first row of the left group), DQ-DETR detected only 40 of the 256 ground truths (SPOT-DETR, 170; RT-DETR, 113). These trends agree with the differences in APvt

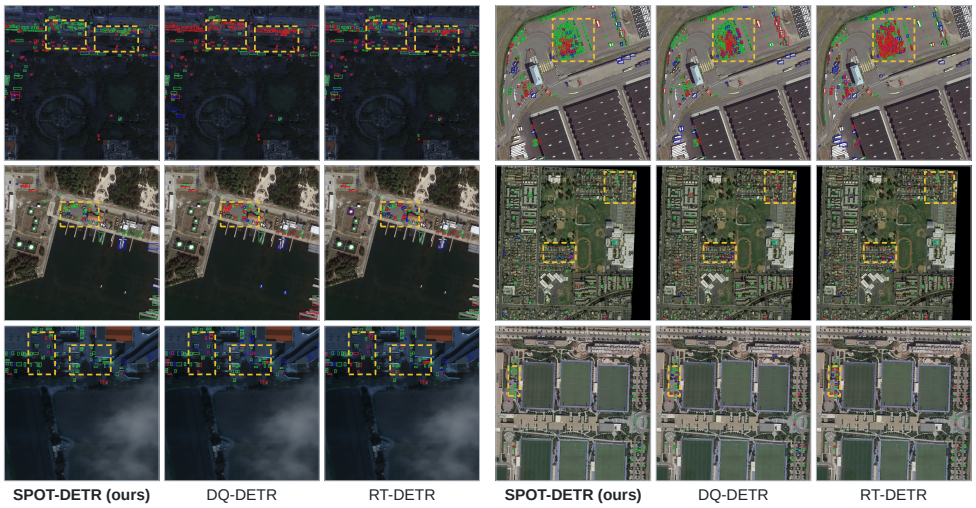


Fig. 5 Comparison of the detection results of SPOT-DETR, DQ-DETR and RT-DETR on the AI-TOD-v2 test split. Green, blue and red boxes are TP, FP and FN, respectively, and yellow dashed boxes are regions where the differences between models are pronounced.

in Table 1 and in the VE AP in Table 3. FPs, however, varied by scene: in the dark road scene, DQ-DETR had fewer FPs (6) than SPOT-DETR (24). Because Fig. 5 shows examples selected to highlight the differences, quantitative conclusions should be drawn from Tables 1 and 3.

4.3 Ablation studies

4.3.1 Effect of combining the components

Table 4 Component ablation on the AI-TOD-v2 test split. All configurations with SGS or ATOM include the X_2 path.

Configuration	AP	APvt	APt	APs
Baseline	0.298	0.157	0.305	0.339
+SGS	0.332	0.207	0.343	0.360
+ATOM	0.338	0.200	0.349	0.363
+SGS +ATOM	0.337	0.212	0.353	0.363

Table 4 shows the results of adding SGS and ATOM to the baseline, separately or together. Because both components operate on the X_2 path, all three configurations other than the baseline include X_2 in the feature pyramid. The AP and APvt of the three configurations were 3.4–4.0% and 4.3–5.5% higher, respectively, than those of the baseline. That this gain appeared regardless of which component was added

suggests that most of the benefit of SPOT comes from keeping in the feature pyramid the spatial cues that are lost in later downsampling.

The two components differed across size ranges. The APvt of +SGS and SPOT-DETR, which include SGS, was 20.7% and 21.2%, respectively, higher than the 20.0% of +ATOM, which does not. SPOT-DETR, which combines both components, had the highest APvt and APt, at 21.2% and 35.3%, respectively, and the same APs as +ATOM (36.3%); its AP of 33.7% was 0.1% lower than that of +ATOM (33.8%), which was trained with $\lambda_{\text{rank}} = 0.50$.

Combining the two components thus raised the very tiny and tiny ranges most while keeping the AP and APs of +ATOM. Table 4 supports the interpretation that the X_2 path forms the common basis of the improvement and that SGS adds a gain in the very tiny range on top of ATOM. We kept both components in the final model because this combination had the highest APvt and APt in Table 4 and, compared with the baseline and $+X_2$ configurations, a higher very tiny recall (Table 5) and a smaller median center error (Section 4.3.2).

Figure 6 compares the behavior of the configurations that add SGS or ATOM to the baseline in the same scenes, which were selected to show the differences clearly. Because both configurations include the X_2 path, the differences in the figure result from adding the X_2 path together with the respective component. In the dark scene of Fig. 6(a), each cell of the P_3 of the baseline is 8 pixels on a side, so the vehicles in the zoom-in region span one or two cells, whereas in +SGS the same vehicles appear as the responses of smaller cells that are 4 pixels on a side. The baseline missed all four vehicles in the zoom-in region, whereas +SGS detected all four, and over the whole image, the number of TPs among the 28 ground truths increased from 21 to 25. In Fig. 6(b), the grid cell centers of the baseline lie at regular intervals regardless of the vehicle positions, whereas the atom centers of +ATOM gather on the vehicles, and atoms with higher occupancy probability lie closer to the vehicle centers. In this scene, +ATOM reduced the misaligned predictions and misses of the baseline on adjacent vehicles: the number of TPs among the 51 ground truths increased from 38 to 44, and the number of FPs decreased from 20 to 6.

4.3.2 Analysis of initial candidates

Table 5 Recall of the top 900 initial candidates (IoU≥0.5) on the AI-TOD-v2 test split.

Model / candidate source	vt	t	s	m
Baseline (P_3–P_5 grid candidates)	0.385	0.845	0.903	0.902
$+X_2$ (P_2–P_5 grid candidates)	0.628	0.912	0.930	0.876
SPOT-DETR (atom + P_3–P_5 grid candidates)	0.671	0.908	0.898	0.855
SPOT-DETR, atom candidates only	0.671	0.908	0.888	0.789
SPOT-DETR, P_3–P_5 grid candidates only	0.000	0.022	0.184	0.535

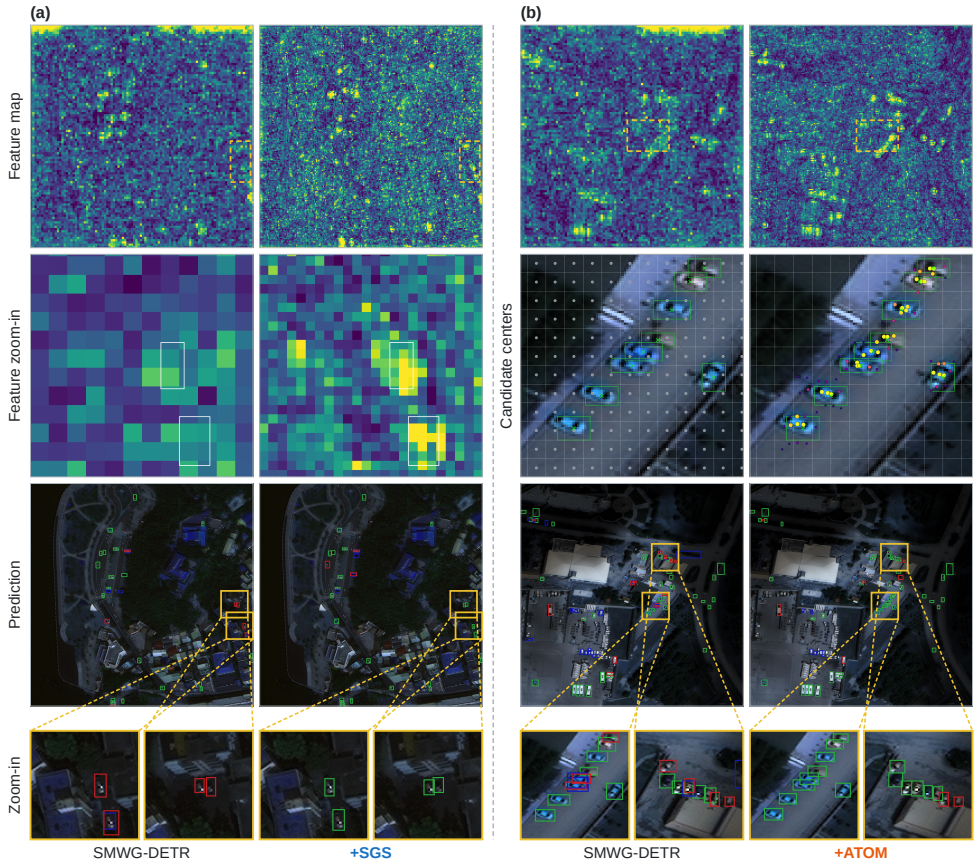


Fig. 6 Examples of the behavior of the configurations with SGS or ATOM. (a) compares SMWG-DETR with +SGS, and (b) compares SMWG-DETR with +ATOM, each pair on the same AI-TOD-v2 test image. The first row shows the channel-averaged absolute value of the finest feature map (P_3 for SMWG-DETR and P_2 resolution for the others). The second row shows the first zoom-in region: (a) the feature map with ground-truth boxes (white), and (b) grid cell centers (gray dots) and atom centers (yellower for higher occupancy probability) together with ground-truth boxes (green) and ground-truth centers (×). The third row shows the detection results, and the fourth row magnifies the gold boxed regions. Green, blue and red boxes are TP, FP and FN, respectively.

Table 5 lists the recall of the top 900 initial candidates by size range. The P_3–P_5 grid candidates of the baseline covered only 38.5% of very tiny ground truths with an IoU of 0.5 or higher, far below the recalls of the tiny (84.5%) and larger ranges. Adding X_2 and also using the grid candidates of P_2 raised the very tiny recall to 62.8%, and SPOT-DETR, which uses atom candidates instead of the grid candidates of P_2, reached 67.1%. In contrast, the tiny, small and medium recalls of SPOT-DETR were 0.4%, 3.2% and 2.1% lower, respectively, than those of the $+X_2$ configuration. Because the two configurations also differ in whether SGS is used, we do not attribute these decreases to the atoms alone. Broken down by source, all of the very tiny recall

of SPOT-DETR came from atom candidates (67.1%), and the P_3–P_5 grid candidates within the top 900 covered almost no very tiny or tiny ground truths (0.0% and 2.2%).

The proportion of atoms in the top 900 varied widely across images, with a mean of 23.9%, a median of 9.5% and a 10th–90th percentile range of 1.7–71.5%. The predicted atom centers deviated from the cell centers by 0.165 cells on average, and 57.6% and 24.8% of them deviated by more than 0.1 and 0.25 cells, respectively, showing that the continuous center coordinates were used (an example for +ATOM is shown in Fig. 6(b)). The median distance between the center of a ground truth with a longer side of 16 pixels or less and the nearest atom was 0.88 pixels, and the proportions within 1, 2 and 4 pixels were 56.7%, 89.9% and 99.8%, respectively. As the occupancy probability bin rose from 0.1–0.2 to 0.9–1.0, the proportion of positive atoms (those with a ground truth with a longer side of 16 pixels or less within 4 pixels) increased monotonically from 2.0% to 48.8%, indicating that the occupancy logit reflected the presence of objects.

For candidates matched to a ground truth with an IoU of 0.25 or higher, the median center error was 1.49 pixels for the grid candidates of the baseline, 1.13 pixels for the grid candidates of the $+X_2$ configuration and 1.00 pixel for the atoms of SPOT-DETR. The center error thus decreased by 0.36 pixels with candidates at P_2 resolution and by a further 0.13 pixels with atoms. Because this further reduction is small, we do not attribute the effect of ATOM on final performance to the off-grid initial centers alone.

The relative magnitude of the raster residual, $|\widetilde{P}_2 - P_2|/|P_2|$, was 0.47% in object cells and 0.18% in background cells. Although the relative magnitude of the raster residual was about 2.6 times larger in object cells than in background cells, it remained below 0.5% of the P_2 feature norm in both; we therefore interpret the raster path as a small complement that does not replace the dense features.

4.3.3 Feature map visualization

Figure 7 compares the finest feature maps of the decoder memory for the four configurations in Table 4. As in [13], the first principal component of PCA is shown for the comparison across configurations. In P_3, the finest feature map of the decoder memory in the baseline, adjacent tiny objects merged into a single response within one or two cells. In contrast, in the P_2 or $\widetilde{P}_2$ of the three configurations that include the X_2 path, the same objects appeared as separate responses, consistent with the higher very tiny candidate recall in Table 5. Although the raster residual is added to P_2 to form the $\widetilde{P}_2$ of +ATOM and SPOT-DETR, its magnitude is below 0.5% of the P_2 feature norm (Section 4.3.2), so the effect of ATOM is more evident in the positions of the initial candidates (Section 4.3.2 and Fig. 6(b)) than in the feature maps.

The differences between the feature maps should be interpreted as differences in spatial resolution and separation rather than in response strength. When SPOT-DETR and the baseline were compared on the same 500 randomly chosen test images, the feature activation contrast between ground-truth interiors and the background over the 498 valid images was 2.083±0.658 for SPOT-DETR and 2.082±0.651 for the baseline, with a paired per-image difference of 0.001±0.336. The learned spatial gate of SGS was also close to its upper bound of 2 in both the foreground and the background, so SGS strengthened the local residual uniformly rather than suppressing background

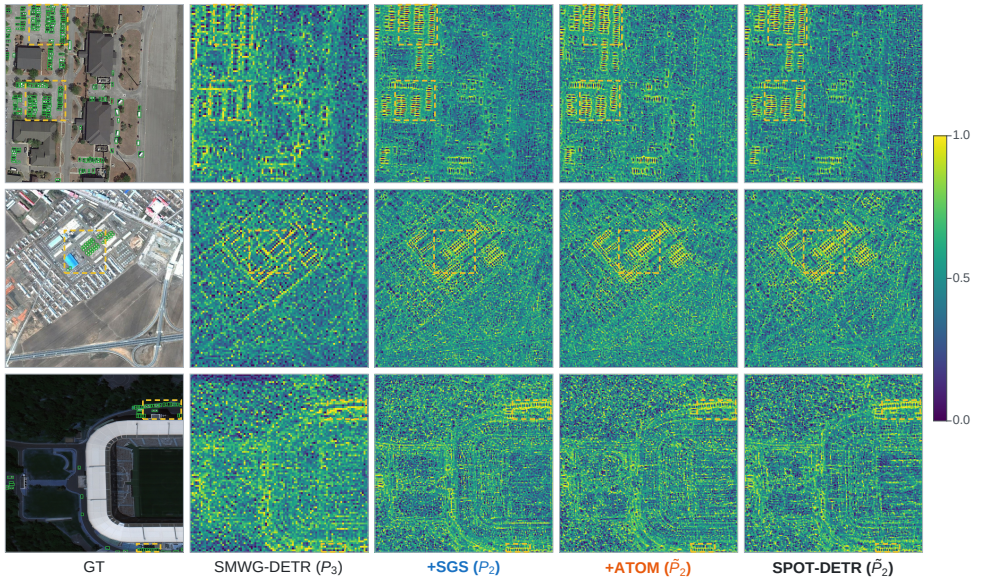


Fig. 7 Finest feature maps of the decoder memory on AI-TOD-v2 test images. The first column shows the ground truth (GT), and the feature maps show the first principal component obtained by principal component analysis (PCA), sign-aligned so that ground-truth regions appear bright, and normalized to 0–1. Yellow dashed boxes are regions where adjacent objects are clustered.

responses position by position. Rather than amplifying object responses relative to the background, SPOT-DETR thus separated, at high resolution, the responses of adjacent objects that merged within one or two cells in the baseline.

4.3.4 Hyperparameter sensitivity

In reduced-scale experiments with 30% of the data and 18 epochs, the AP for the atom candidate allocation ratio $q \in \{0.25, 0.50, 0.75, 1.00\}$ was 21.0%, 20.3%, 19.8% and 20.8%, respectively, and the AP for $\lambda_{\text{rank}} \in \{0.25, 0.50, 0.75\}$ was 20.9%, 20.8% and 20.9%. For longer-side thresholds of 12, 16 and 20 pixels for the auxiliary supervision, the AP was 20.1%, 20.8% and 20.6%. The AP varied by at most 1.2% across the tested values of each hyperparameter, with the largest variation for q and the smallest for λ_{rank}. Because these results were obtained under reduced conditions, they indicate trends rather than the optimal values on the full data.

4.4 Efficiency analysis

Table 6 lists the number of parameters, giga floating-point operations (GFLOPs), frames per second (FPS) and latency of each detector, measured with an 800×800 input. SPOT-DETR has 58.28M parameters and 382.4 GFLOPs, which are 7.80M and 235.1 GFLOPs more, respectively, than those of the SMWG-DETR baseline. Because SPOT processes X_2 and P_2, which the baseline does not, the computational cost increases substantially. For an 800×800 input, P_2 is 200×200, four times the size

Table 6 Computational efficiency comparison on AI-TOD-v2. The AP of SPOT-DETR is the mean ± standard deviation over three seeds. Red and blue indicate the best and second-best results in each column, respectively.

Method	Params (M)↓	GFLOPs↓	FPS↑	Latency (ms)↓	AP↑
CNN-based					
DNTR [20]	128.82	449.6[3]	0.96	1045.8	0.215
ORFENet [21]	34.01	372.9	17.17	58.3	0.251
LTDNet [22]	4.87	38.7	29.56	33.8	0.230
DETR-based					
DINO [11]	47.55	179.9	16.55	60.4	0.279
RT-DETR [15]	42.74	112.8	29.20	34.2	0.296
RT-DETRv2 [35]	36.45	81.1	35.70	28.0	0.279
DQ-DETR [12]	58.93	910.0[1]	5.34	187.1	0.302
Dome-DETR [14]	13.23	86.4[1]	2.78[2]	360.2	0.262
SMWG-DETR [13]	50.48	147.3	18.31	54.6	0.298
SPOT-DETR (ours)	58.28	382.4	9.14	109.4	0.333 ± 0.003

of P_3; ATOM generates 20,000 atoms from 10,000 blocks; and the decoder memory retains 53,125 tokens of $\widetilde{P}_2$ and P_3–P_5. Even though local rasterization is limited to a neighborhood of at most 3×3 cells per atom, these additions reduced the throughput of SPOT-DETR to 9.14 FPS (latency of 109.4 ms), about half that of SMWG-DETR (18.31 FPS, 54.6 ms).

Nevertheless, SPOT-DETR had a shorter latency than DQ-DETR (187.1 ms) and Dome-DETR (360.2 ms), which are DETR-based detectors for tiny objects, and achieved a 3.1% higher AP than DQ-DETR with a similar number of parameters (58.93M). On the other hand, it is slower than detectors suited to real-time processing, such as RT-DETRv2 (35.70 FPS) and LTDNet (29.56 FPS). SPOT-DETR is therefore better suited to wide-area image analysis, in which the detection accuracy for tiny objects matters more than processing speed.

4.5 Limitations and failure cases

Although SPOT-DETR improved tiny object detection overall, misses and false detections still occur in scenes such as those in Fig. 8. In extremely dense scenes in which the number of objects far exceeds the number of queries (900), as in Fig. 8(a), most objects are missed. A DETR-based detector that predicts a fixed number of queries cannot detect more objects in an image than it has queries. In the second image of Fig. 8(a), all 900 predictions exceeded the score threshold, yet 853 of the 1,452 ground truths remained undetected, and when the density was extreme, as in the first image with 2,667 ground truths, even the number of predictions exceeding the score threshold dropped sharply. Test images with more than 900 objects account for only 0.08% of the test split but contain 3.8% of all objects. Combining SPOT-DETR with methods that adjust the number of queries according to object density [12, 14] could address

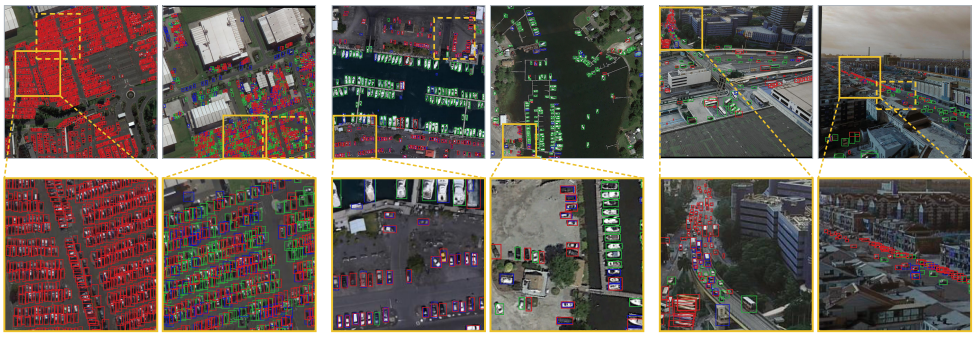


Fig. 8 Failure cases of SPOT-DETR. (a) Extremely dense scenes in which the number of objects exceeds the number of queries (900), (b) scenes that mix look-alike categories such as ships and vehicles and (c) scenes captured obliquely by drones. Green, blue and red boxes are TP, FP and FN, respectively, and yellow dashed boxes are regions where failures are pronounced. The bottom row magnifies the gold boxed regions of the images above.

such scenes. In harbors where ships and vehicles coexist, as in Fig. 8(b), the two look-alike categories were confused with each other, producing FPs and FNs at the same locations. At a size of a few pixels, the two categories have nearly the same shape and brightness, so the boundary and local contrast cues that SPOT preserves are not enough on their own to distinguish them. This is a fundamental difficulty of tiny object detection, in which appearance information is extremely scarce, and it could be reduced by also exploiting the surrounding context, such as water surfaces and parking areas. In scenes captured obliquely by drones, as in Fig. 8(c), distant objects become smaller with perspective and overlap one another, so the size and spacing of objects vary greatly even within a single image. Because we did not use data augmentation (Section 4.1.3), augmentation that reflects perspective-induced changes in size could reduce misses in such scenes.

In addition, the high-resolution feature path makes SPOT-DETR run at about half the speed of the baseline (Section 4.4), which limits its use in applications that require real-time processing. Concentrating high-resolution computation on foreground regions is one way to reduce this cost.

5 Conclusion

We analyzed two factors that limit DETR-based detectors in tiny object detection in remote sensing images: the loss of fine details caused by downsampling and the localization error of initial candidates fixed at cell centers. To address both on a single high-resolution feature path, we proposed SPOT and built SPOT-DETR on it. SPOT includes X_2 in the feature pyramid and refines it with SGS, and it uses the off-grid atoms generated by ATOM from the fused P_2 for both the initial queries and the decoder memory. While keeping the L_1–GIoU regression and matching of the baseline unchanged, SPOT-DETR achieved an AP of 37.6% on AI-TOD and 33.3% on AI-TOD-v2, outperforming all compared detectors. On AI-TOD-v2, the largest gain over

the baseline was 5.1% in APvt, and the initial-candidate recall for very tiny objects rose from 38.5% to 67.1%. The component analysis showed that the X_2 path formed the common basis of the improvement, that the configurations with SGS had a higher APvt and that the atoms provided all of the very tiny initial candidates of SPOT-DETR with smaller center errors than the grid candidates of the baseline. Because SPOT-DETR processes high-resolution features, however, it runs at about half the speed of the baseline. In future work, we plan to reduce the computational cost of the high-resolution feature path while maintaining detection performance and to extend the method to images from diverse imaging conditions and other sensors.

Data availability

The AI-TOD and AI-TOD-v2 datasets analyzed in this study are available from the original authors' public repository (https://github.com/jwwangchn/AI-TOD) and project page (https://chasel-tsui.github.io/AI-TOD-v2/), respectively [1, 2]. No new datasets were generated.

References

[1] Wang, J., Yang, W., Guo, H., Zhang, R. & Xia, G.-S. *Tiny object detection in aerial images.* In *2020 25th International Conference on Pattern Recognition (ICPR)*, 3791–3798 (IEEE, 2021).

[2] Xu, C. *et al.* Detecting tiny objects in aerial images: A normalized Wasserstein distance and a new benchmark. *ISPRS Journal of Photogrammetry and Remote Sensing* **190**, 79–93 (2022).

[3] Lin, T.-Y. *et al. Feature pyramid networks for object detection.* In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 936–944 (IEEE, 2017).

[4] Yang, C., Huang, Z. & Wang, N. *QueryDet: Cascaded sparse query for accelerating high-resolution small object detection.* In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 13658–13667 (IEEE, 2022).

[5] Yang, F., Fan, H., Chu, P., Blasch, E. & Ling, H. *Clustered object detection in aerial images.* In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, 8310–8319 (IEEE, 2019).

[6] Li, C., Yang, T., Zhu, S., Chen, C. & Guan, S. *Density map guided object detection in aerial images.* In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 737–746 (IEEE, 2020).

[7] Carion, N. *et al. End-to-end object detection with transformers.* In *Computer Vision – ECCV 2020*, Vol. 12346 of *Lecture Notes in Computer Science*, 213–229 (Springer, 2020).

[8] Zhu, X. *et al. Deformable DETR: Deformable transformers for end-to-end object detection.* In *International Conference on Learning Representations* (2021). URL https://openreview.net/forum?id=gZ9hCDWe6ke.

[9] Liu, S. *et al. DAB-DETR: Dynamic anchor boxes are better queries for DETR.* In *International Conference on Learning Representations* (2022). URL https://openreview.net/forum?id=oMI9PjOb9Jl.

[10] Li, F. *et al. DN-DETR: Accelerate DETR training by introducing query DeNoising.* In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 13609–13617 (IEEE, 2022).

[11] Zhang, H. *et al. DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection.* In *International Conference on Learning Representations* (2023). URL https://openreview.net/forum?id=3mRwyG5one.

[12] Huang, Y.-X., Liu, H.-I., Shuai, H.-H. & Cheng, W.-H. *DQ-DETR: DETR with dynamic query for tiny object detection.* In *Computer Vision – ECCV 2024*, Vol. 15134 of *Lecture Notes in Computer Science*, 290–305 (Springer, 2025).

[13] Chen, M. *et al.* SMWG-DETR: DETR enhanced by Fourier spectral modulation and wavelet-guided fusion for tiny object detection. *IEEE Transactions on Geoscience and Remote Sensing* **64**, 1–17 (2026). Art. no. 5605117.

[14] Hu, Z. *et al. Dome-DETR: DETR with density-oriented feature-query manipulation for efficient tiny object detection.* In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, 101–110 (ACM, 2025).

[15] Zhao, Y. *et al. DETRs beat YOLOs on real-time object detection.* In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 16965–16974 (IEEE, 2024).

[16] Xu, C., Wang, J., Yang, W. & Yu, L. *Dot distance for tiny object detection in aerial images.* In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 1192–1201 (IEEE, 2021).

[17] Xu, C. *et al. RFLA: Gaussian receptive field based label assignment for tiny object detection.* In *Computer Vision – ECCV 2022*, 526–543 (Springer, 2022).

[18] Rezatofighi, H. *et al. Generalized intersection over union: A metric and a loss for bounding box regression.* In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 658–666 (IEEE, 2019).

[19] Yuan, X., Cheng, G., Yan, K., Zeng, Q. & Han, J. *Small object detection via coarse-to-fine proposal generation and imitation learning.* In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, 6294–6304 (IEEE, 2023).

[20] Liu, H.-I. *et al.* A DeNoising FPN with Transformer R-CNN for tiny object detection. *IEEE Transactions on Geoscience and Remote Sensing* **62**, 1–15 (2024). Art. no. 4704415.

[21] Liu, D., Zhang, J., Qi, Y., Wu, Y. & Zhang, Y. Tiny object detection in remote sensing images based on object reconstruction and multiple receptive field adaptive feature enhancement. *IEEE Transactions on Geoscience and Remote Sensing* **62**, 1–13 (2024). Art. no. 5616213.

[22] Liu, D., Zhang, J., Qi, Y., Xi, Y. & Jin, J. Exploring lightweight structures for tiny object detection in remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing* **63**, 1–15 (2025). Art. no. 5623215.

[23] Yang, J., Yue, X. & Wu, L. A collaborative multi-attention network for real-time small object detection in UAV imagery. *Scientific Reports* **16**, 5852 (2026).

[24] Liu, S. *et al.* A dynamic element-activated non-semantic sparse attention method for remote sensing small object detection. *Scientific Reports* **16**, 11577 (2026).

[25] Liu, H. & Su, T. Aero-LiteNet: Robust aerial small object detection via multi-scale fusion and neighborhood-aware optimization. *Scientific Reports* **16**, 17389 (2026).

[26] He, K., Zhang, X., Ren, S. & Sun, J. *Deep residual learning for image recognition.* In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778 (IEEE, 2016).

[27] Yu, W. *et al. MetaFormer is actually what you need for vision.* In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 10809–10819 (IEEE, 2022).

[28] Yu, W. *et al.* MetaFormer baselines for vision. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **46**, 896–912 (2024).

[29] Lin, T.-Y., Goyal, P., Girshick, R., He, K. & Dollár, P. *Focal loss for dense object detection.* In *2017 IEEE International Conference on Computer Vision (ICCV)*, 2999–3007 (IEEE, 2017).

[30] Zhang, H., Wang, Y., Dayoub, F. & Sünderhauf, N. *VarifocalNet: An IoU-aware dense object detector.* In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 8510–8519 (IEEE, 2021).

[31] Paszke, A. *et al. PyTorch: An imperative style, high-performance deep learning library.* In *Advances in Neural Information Processing Systems 32*, 8026–8037 (2019). URL https://proceedings.neurips.cc/paper_files/paper/2019/file/bdbca288fee7f92f2bfa9f7012727740-Paper.pdf.

[32] Chen, K. *et al.* MMDetection: Open MMLab detection toolbox and benchmark. *arXiv preprint arXiv:1906.07155* (2019).

[33] Deng, J. *et al. ImageNet: A large-scale hierarchical image database.* In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, 248–255 (IEEE, 2009).

[34] Loshchilov, I. & Hutter, F. *Decoupled weight decay regularization.* In *International Conference on Learning Representations* (2019). URL https://openreview.net/forum?id=Bkg6RiCqY7.

[35] Lv, W. *et al.* RT-DETRv2: Improved baseline with bag-of-freebies for real-time detection transformer. *arXiv preprint arXiv:2407.17140* (2024).

Author contributions. Conceptualization and methodology, K.E., C.P. and H.J.; software, data curation and visualization, K.E.; validation, formal analysis and investigation, K.E. and C.P.; writing—original draft preparation, K.E.; writing—review and editing, C.P., E.K. and H.J.; supervision, project administration and funding acquisition, E.K. and H.J. All authors have read and agreed to the published version of the manuscript.

Funding. This work was supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (No. RS-2026-25617952, AI Star Fellowship Support).

Declarations.

Competing interests. The authors declare no competing interests.

Ethical approval. This study was conducted using publicly available datasets and did not involve any new human participants or the collection of private human data; therefore, ethics committee or institutional review board approval was not required.